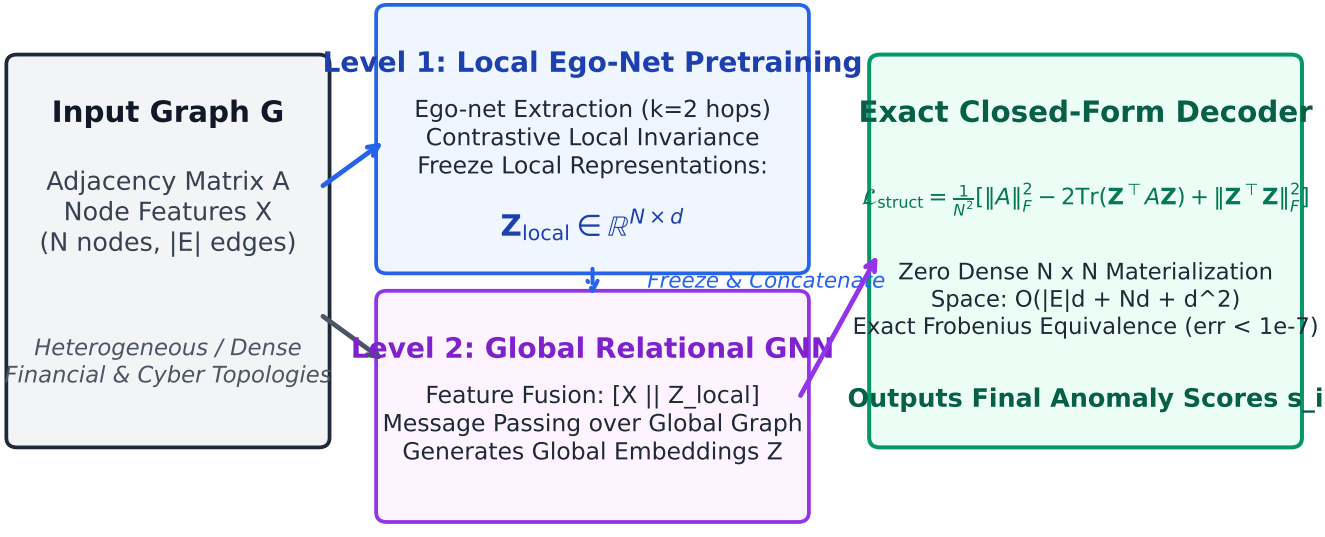
Input Graph G
Adjacency Matrix A
Node Features X
(N nodes, |E| edges)
Heterogeneous / Dense
Financial & Cyber Topologies
Level 1: Local Ego-Net Pretraining
Ego-net Extraction (k=2 hops)
Contrastive Local Invariance
Freeze Local Representations:
$\mathbf{Z}_{\text{local}} \in \mathbb{R}^{N \times d}$
Freeze & Concatenate
Level 2: Global Relational GNN
Feature Fusion: [X || Z_local]
Message Passing over Global Graph
Generates Global Embeddings Z
Exact Closed-Form Decoder
$\mathcal{L}_{\text{struct}} = \frac{1}{N^2}[\|A\|_F^2 - 2\text{Tr}(\mathbf{Z}^\top A\mathbf{Z}) + \|\mathbf{Z}^\top\mathbf{Z}\|_F^2]$
Zero Dense N x N Materialization
Space: O(|E|d + Nd + d^2)
Exact Frobenius Equivalence (err < 1e-7)
Outputs Final Anomaly Scores s_i